

Kawasaki Disease Guidelines

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A) Diagnostic Criteria for Kawasaki Disease (KD)¹

Fever for more than 5 days with ≥ 4 of the following clinical signs not explained by another disease process:

1. Bilateral non-suppurative conjunctival injection
2. Changes in oropharyngeal mucous membranes – ≥ 1 of:
 - a. Injected and/or fissured lips
 - b. Strawberry tongue
 - c. Injected pharynx
3. Cervical lymphadenopathy with at least one node ≥ 1.5 cm
4. Changes in peripheral extremities, including:
 - a. Erythema and/or edema of hands and feet (acute phase)
 - b. Periungual desquamation (convalescent phase)
5. Polymorphous rash – primarily truncal, non-vesicular

Points to note:

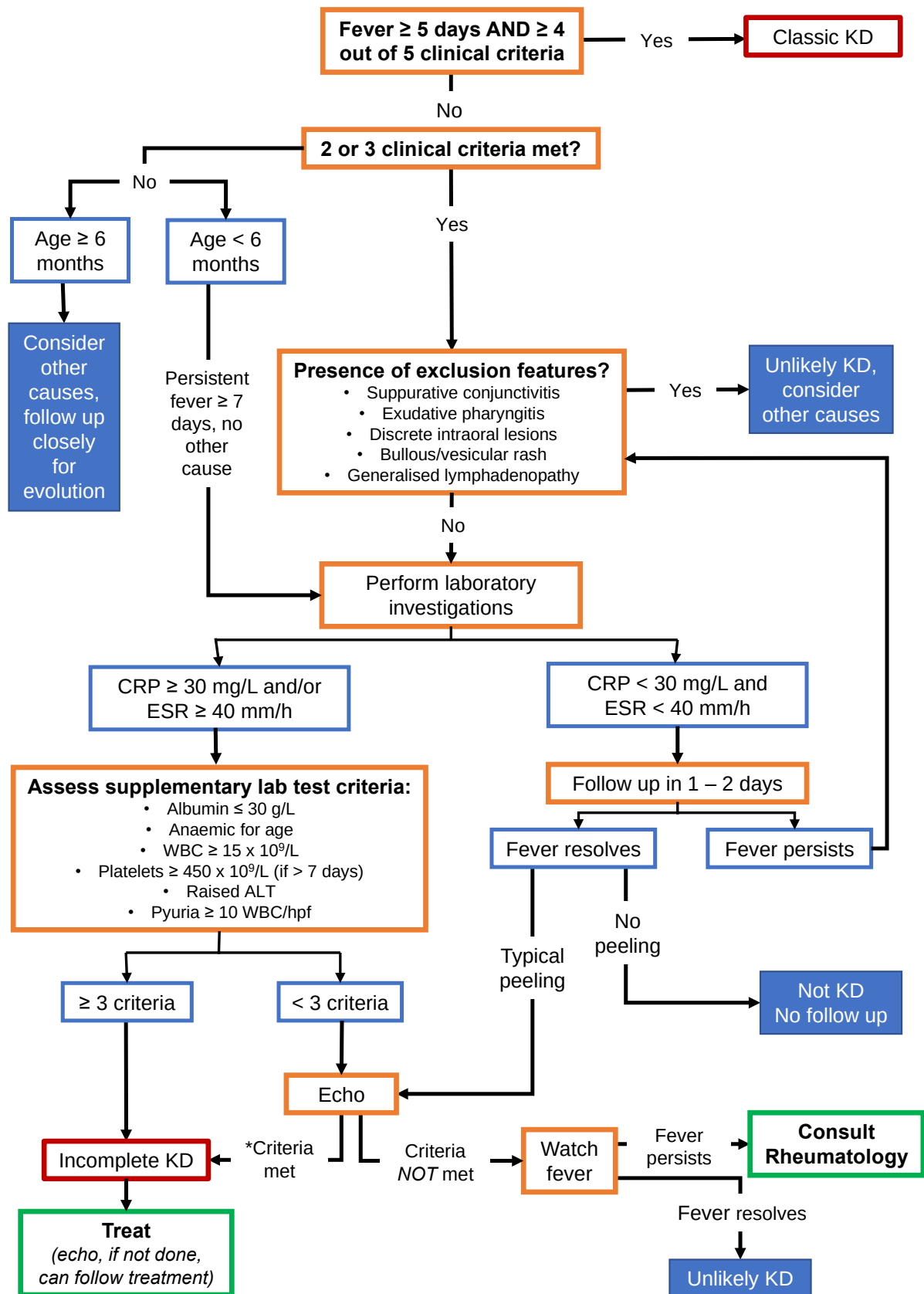
- **Fewer than 4 out of 5 criteria is required if echocardiogram shows coronary artery involvement.**
- KD can be diagnosed in the presence of classic features even though fever is < 5 days of duration
- *Clinical features to support the diagnosis of KD² can be found in [Appendix 1](#).*

B) Atypical or Incomplete KD

Atypical or incomplete KD refers to children who do not meet the above diagnostic criteria. Children less than 1 year of age are more likely to present with atypical or incomplete KD.

Recommendations³:

- Any child **< 6 months** of age with **persistent fever ≥ 7 days and evidence of inflammation** (CRP ≥ 30 mg/L and ESR ≥ 40 mm/h), require an echocardiogram (*even in absence of positive clinical criteria*).



Adapted from "Kawasaki Disease: What to do with Incomplete Cases?" by JH Baumer, 2005, Arch Dis Child Educ Pract Ed, 90, p103.

*Echocardiogram is positive if any one of three criteria is/are met:

- 1) Z-score of left anterior descending coronary artery (LAD) or right coronary artery (RCA) is ≥ 2.5
- 2) Presence of coronary artery aneurysm
- 3) ≥ 3 suggestive features present:
 - Perivascular brightness
 - Lack of tapering
 - Decreased left ventricular function
 - Mitral regurgitation
 - Pericardial effusion
 - Z-scores in LAD or RCA of 2 – 2.5

C) Investigations

1. Full blood count (FBC)
2. Erythrocyte sedimentation rate (ESR)
3. C-reactive protein (CRP)
4. Liver function test (LFT)
5. Renal panel
6. Urine dipstick/urine microscopy/urine culture: sterile pyuria (if diagnosis of KD in doubt)
 - a. *Note that bagged sample will distinguish between UTI and urethritis in KD.*
7. Electrocardiogram (ECG)
8. Echocardiogram: Access to echocardiogram should **not** preclude treatment for KD.
 - a. *Non-urgent echocardiogram should be done at time of diagnosis*
 - b. *Earlier if:*
 - i. *there is haemodynamic instability, or*
 - ii. *diagnosis is in doubt*

Other investigations to consider in specific circumstances:

9. Ferritin (*if suspecting macrophage activation syndrome or Multisystem Inflammatory Syndrome in Children, MIS-C*)
10. Cardiac enzymes (CK, CKMB, troponin I) and pro-BNP (*if suspecting MIS-C or if haemodynamic instability*)
11. DIC screen (*if suspecting MIS-C*)
12. COVID-19 serologies (*if suspecting MIS-C*)

D) Treatment

I) Intravenous Immunoglobulin (IVIg) 2g/kg⁴

- Reduces incidence of coronary artery aneurysm (CAA) from 25% to 5% in typical KD.
- Response to IVIg is defined as resolution of fever ($T < 37.5^{\circ}\text{C}$) and mucocutaneous features. This may take up to 36 hours from end of IVIg infusion.
- 20-40% of children do not respond to first dose of IVIg⁵.
- If diagnosis of KD is made 10 days after onset of fever, IVIg should be given if the patient continues to be febrile or have signs of persistent inflammation (e.g. raised inflammatory markers)⁶.

II) Aspirin (ASA)

- Anti-inflammatory (moderate) dose **30 - 50 mg/kg/day** in 3 divided doses⁷
 - High-dose aspirin, combined with IVIg, does not confer additional benefit for CAA outcomes⁵
- Consider oral famotidine or omeprazole for gastric protection.
- Reduce to anti-platelet dose **3 – 5 mg/kg daily**
 - Once afebrile for ≥ 48 hours and inflammation has subsided OR
 - If on concomitant steroids.

This dose is usually discontinued at 6 – 8 weeks from onset of symptoms if echocardiogram remains normal. For children who develop coronary abnormalities, aspirin may be continued indefinitely.

- Concomitant use of ibuprofen antagonizes the irreversible platelet inhibition induced by aspirin and should generally be avoided in children with coronary artery aneurysms⁸.

III) Corticosteroids

There is increasing evidence that the addition of steroids to IVIg and aspirin in the primary therapy of KD lowers the prevalence of coronary artery abnormalities, duration of fever and inflammation but this is predicated on the need for accurate predictive scoring systems to identify children at high risk of IVIg failure⁹.

A small study in Singapore showed that none of the available scoring systems achieve high sensitivity in our local population, but that the Fukunishi, Egami and Kobayashi scoring systems may be the more reliable ones until further evidence becomes available¹⁰.

Indications for early corticosteroid use in addition to (first) IVIg^{5,11}:

1. Infants < 6 months
2. Presence of CAA, coronary artery dilatation (z-score ≥ 2.5), other cardiac manifestations (i.e. mitral regurgitation, pericardial effusion), while still febrile.
3. KD with complications of shock or macrophage activation syndrome.

Consider adding on steroids to second IVIg in the event of IVIg resistance, especially in the setting of significant ongoing systemic inflammation or with coronary artery abnormalities.

Consult Rheumatology before starting corticosteroids.

- Any of the following corticosteroid regimens may be used:
 - 1 IV methylprednisolone 0.8 mg/kg BD x 5 – 7 days OR until CRP normalizes *then*, PO prednisolone 2 mg/kg/day and wean over next 2-3 weeks
 - 2 PO prednisolone 2 mg/kg/day for 5 – 7 days OR until CRP normalizes *then*, wean over next 2-3 weeks
 - 3 IV methylprednisolone 10-30 mg/kg daily x 3 days *then*, PO prednisolone 2 mg/kg/day until day 7 OR until CRP normalizes; wean over next 2-3 weeks

IV) Infliximab (6-10 mg/kg) if there is ongoing systemic inflammation despite 2 doses of IVIg and systemic corticosteroids.

E) Rheumatology Referral should be considered when:

- Diagnosis of KD is in doubt
- Corticosteroids indicated (see above)
- Failed first dose of IVIg
- Complications of KD – KD Shock Syndrome; Macrophage Activation Syndrome

F) Response to treatment is defined as

1. resolution of fever ($T < 37.5^{\circ}\text{C}$) AND
2. resolution of mucocutaneous features AND
3. normalization of CRP ($< 10 \text{ mg/L}$) or halving of CRP every 24 hours.

This may take up to 36 hours from end of IVIg infusion.

What to do if fever ($T \geq 37.5^{\circ}\text{C}$) persists or recurs ≥ 36 hours after the end of IVIg?

1. Assess child for possible causes of persistent fever (e.g. thrombophlebitis, new infection)
2. Repeat inflammatory markers: FBC, CRP
3. Give second dose of 2g/kg IVIg with strong consideration for corticosteroids as well^{5,11}.

What to do if fever resolves but mucocutaneous signs are still present ≥ 36 hours after the end of IVIg?

Refer Rheumatology as consideration for ongoing treatment must be made.

G) Follow up

I) Investigations & Clinic review

Before discharge	10 – 14 days post-discharge	6 – 8 weeks from initial presentation
FBC, CRP	Cardiology outpatient:	Cardiology outpatient:

II) Immunizations

- The following time interval between IVIg and vaccinations is recommended¹²:
 - Inactivated** vaccinations: no interval required
 - Live** vaccinations (MMR, varicella): **≥ 11 months** after IVIg
 - Provide parents with memo ([Appendix 2](#)) to defer live vaccines during this period
- If MMR/varicella is administered within 14 days prior to administration of IVIg, revaccinate.

Source: Drs Pauline Chan, Ooi Pei Ling, Elizabeth Ang (July 2017) with Prof Quek Swee Chye

Updated by Drs Pauline Chan and Elizabeth Ang (July 2023)

Next review due: 2026

This information aims to guide you in the management of children's diseases and reflects current practice in the KTP-National University Children's Medical Institute. Considerable care has been taken to ensure that the protocols are accurate but they are not a substitute for asking advice from those more familiar with the problems patients present. If any aspect of patient care is unclear, consult a more senior or subspecialist doctor.

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APPENDIX 1: Supportive Features for KD²

A. Cardiovascular findings

- i. Congestive heart failure, myocarditis, pericarditis, or valvular regurgitation
- ii. Coronary artery abnormalities
- iii. Aneurysms of medium-sized non-coronary arteries
- iv. Raynaud phenomenon
- v. Peripheral gangrene

B. Musculoskeletal system

- i. Arthritis
- ii. Arthralgia

C. Gastrointestinal tract

- i. Diarrhea, vomiting, abdominal pain
- ii. Hepatic dysfunction
- iii. Hydrops of the gallbladder

D. Respiratory tract

- i. Pulmonary nodules and interstitial infiltrates
- ii. Pleural effusion

E. Central nervous system

- i. Extreme irritability
- ii. Aseptic meningitis
- iii. Peripheral facial nerve palsy
- iv. Sensorineural hearing loss

F. Genitourinary system

- i. Urethritis/meatitis
- ii. Testicular swelling

G. Other findings

- i. Erythema and induration at BCG site**
- ii. Anterior uveitis
- iii. Desquamating rash in groin**

H. Lab findings in acute Kawasaki disease (KD)

- i. Neutrophilia with immature forms**
- ii. Elevated erythrocyte sedimentation rate (ESR)**
- iii. Elevated C-reactive protein (CRP) level**
- iv. Elevated serum α 1-antitrypsin level
- v. Anaemia**
- vi. Abnormal plasma lipids
- vii. Hypoalbuminemia**
- viii. Thrombocytosis after first week of illness
- ix. Sterile pyuria**
- x. Elevated serum transaminases**
- xi. Pleocytosis of cerebrospinal fluid
- xii. Leukocytosis in synovial fluid

**These features can be easily looked for in the initial clinical and laboratory assessment*

APPENDIX 2: Memo to defer live vaccines for 11 months after IVIg

(Date)

To:
Doctor in charge

Dear Colleague:

RE: (Patient details)

The abovementioned patient received intravenous immunoglobulins (IVIg) for Kawasaki Disease on (date) . He/She should **defer all live vaccinations until 11 months after IVIg**.

He/She **can receive all inactivated vaccinations** after IVIg.

Thank you.

Yours sincerely,

(Doctor's name)

(Designation)

Khoo Teck Puat-National Children's Medical Institute
National University Hospital, Singapore